

Serotonin and Melatonin

Synthesis, transport, receptors and degradation of serotonin (TPH2, SLC6A4, HTR1A/1B/2A/2C, MAOA) and production, receptors and clock of melatonin (MTNR1A/1B, AANAT, ASMT, CLOCK, ARNTL, PER2)

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

What was seen. 46 SNPs were read in 27 genes (31 SNPs of serotonin and 15 of melatonin / clock): 36 of the chip, 8 imputed with $DR2 \geq 0.8$ and 2 not found (both from **ASMT**, in the pseudoautosomal region). As a rule of Sérgio F, the results of neurotransmitters are descriptive, without judgment of good or bad.

Serotonin. There is a coherent pattern of Reduced Receptors: **HTR2A** (rs6313 G/G and rs6311 C/C, Reduced in both), **HTR2C** (rs518147 G and rs3813929 C, Reduced in X) and **HTR1B** (rs130060 A/A Reduced). **HTR1A** (rs6295 C/G) and **HTR1B** rs6296/rs11568817 are mixed. **HTR4** went from "not found" to Altered in both SNPs (Imputed, $DR2$ 1.00). On the production side, **TPH2** (rs4570625 G/T) has 1 allele Increased, and **DDC** has 1 SNP reduced in 3. The vesicle storage has the **SLC18A1** Reduced and the **SLC18A2** Reduced in a SNP (rs363371 G/G) and Typical in the other two. The reuptake (SLC6A4) is contradictory: rs2129785 T/T points to the "short" allele of 5-HTTLPR and rs11867581 G/G for the "long". The degradation (**MAOA**, **ALDH2**) and the modulators **GSK3B** and **HTR6** are Typical. **BDNF** Val66Met is heterozygous (C/T: 1 Typical + 1 Reduced).

Melatonin and clock. The production (**AANAT**) is Typical and **ASMT** has not been read. The receivers are mixed: **MTNR1A** with 2 SNPs Higher and 1 Lower, and **MTNR1B** rs10830963 C/G (1 allele G, the later release of melatonin and glucose in higher fasting). In the clock, **PER2** is Increased, **ARNTL** has 2 SNPs Altered and 1 Typical, the **CLOCK** is mixed and the **CRY2** is Normal. In the elimination, **CYP1A2** has 1 allele Reduced (rs762551 C/A, *1F heterozygous) and **SULT1A1** is Increased.

What does it mean? They are small and added, non-diagnoses. The descriptive portrait is of a route of serotonin with Reduced post-synaptic signaling (**HTR2A**, **HTR2C**, **HTR1B**) and unclarified transport (**SLC6A4**), and of a clock with variants that request regular schedules (**ARNTL**, **CLOCK**, **MTNR1B**). What most modifies these pathways is known: morning light, regular schedules, exercise, sleep and omega-3.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold, or outside the imputed zone, SNP is treated as SNP not found in the data.
- **Variants:** 46 SNPs: 36 of the chip, 8 imputed and 2 not found. All genotypes were reread directly in the raw DNA and in the imputation for this report.
- **Strand:** All genotypes are in the GRCh38 strand +. 0 SNPs needed conversion to complementary strand and 9 are palindromic (A/T or C/G). When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Chromosome X:** Sérgio F has only one copy of X outside the pseudoautosomal regions, so the SNPs of X appear with a single letter (hemizygous).
- **No good/bad judgment:** by Sérgio F rule for neurotransmitters, variants are not classified as favourable or unfavourable. The activity tags of the original report are used (in English: Increased, Reduced, Typical, Altered, Normal, Higher, Lower). The pathways are read as altered (different from Typical/Normal) or typical.
- **Heterozygous:** show the two alleles and the effect of each one; no new label is invented.

3. SNP genotypes at SNP

For each SNP: genotype (strand +; the SNPs of the X with one letter), the effect of each allele according to the original report, the global activity, the health theme and the Enhancers (potentiators) and Blockers (blockers) indicated. When the report does not indicate any, it appears "—".

Serotonin pathway

Biological order: synthesis (TPH2 → 5-HTP → DDC → serotonin), storage in vesicles (SLC18A1/VMAT1, SLC18A2/VMAT2), reuptake (SLC6A4/SERT), receptors (HTR1A, HTR1B, HTR1D, HTR2A, HTR2C, HTR3A, HTR4, HTR6), degradation (MAOA, ALDH2) and modulators (BDNF, GSK3B). The X chromosome SNPs (MAOA, HTR2C) have only one letter.

Genotype and activity

Gene	SNP	Genotype	1 allele → effect	2 allele → effect	Global activity
BDNF	rs6265	C/T	C → Typical activity	T → Reduced activity	Mixed: 1 Typical activity + 1 Reduced activity
MAOA	rs6323	T (hemizygous, X)	T → Typical activity	— (1 copy: X chromosome)	Typical activity (1 copy, X)
MAOA	rs909525	T (hemizygous, X)	T → Typical activity	— (1 copy: X chromosome)	Typical activity (1 copy, X)
ALDH2	rs671	G/G	G → Typical activity	G → Typical activity	Typical activity (2 copies)
GSK3B	rs334558	G/G	G → Typical activity	G → Typical activity	Typical activity (2 copies)
GSK3B	rs6438552	G/G	G → Typical activity	G → Typical activity	Typical activity (2 copies)
GSK3B	rs3755557	T/T	T → Typical activity	T → Typical activity	Typical activity (2 copies)
HTR6	rs1805054	C/C	C → Typical HTR6A activity	C → Typical HTR6A activity	Typical HTR6A activity (2 copies)
HTR1A	rs6295	C/G	C → Reduced activity	G → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
HTR4	rs1345697	C/C	C → Altered activity	C → Altered activity	Altered activity (2 copies)
HTR4	rs6889822	A/A	A → Altered activity	A → Altered activity	Altered activity (2 copies)
HTR2C	rs518147	G (hemizygous, X)	G → Reduced activity	— (1 copy: X chromosome)	Reduced activity (1 copy, X)
HTR2C	rs3813929	C (hemizygous, X)	C → Reduced activity	— (1 copy: X chromosome)	Reduced activity (1 copy, X)
SLC6A4	rs2129785	T/T	T → A=More likely to carry the "short" 5-HTTLPR allele / G=More likely to carry the "long" 5-HTTLPR allele	T → A=More likely to carry the "short" 5-HTTLPR allele / G=More likely to carry the "long" 5-HTTLPR allele	A=More likely to carry the "short" 5-HTTLPR allele / G=More likely to carry the "long" 5-HTTLPR allele (2 copies)
SLC6A4	rs11867581	G/G	G → C=More likely to carry the "long" 5-HTTLPR allele / T=More likely to carry the "short" 5-HTTLPR allele	G → C=More likely to carry the "long" 5-HTTLPR allele / T=More likely to carry the "short" 5-HTTLPR allele	C=More likely to carry the "long" 5-HTTLPR allele / T=More likely to carry the "short" 5-HTTLPR allele (2 copies)
HTR1B	rs6296	C/G	C → Increased activity	G → Reduced activity	Mixed: 1 Enhanced Activity + 1 Reduced Activity
HTR1B	rs130060	A/A	A → Reduced activity	A → Reduced activity	Reduced activity (2 copies)
HTR1B	rs11568817	A/C	A → Reduced activity	C → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
SLC18A2	rs363276	C/C	C → Typical activity	C → Typical activity	Typical activity (2 copies)
SLC18A2	rs363387	T/T	T → Typical activity	T → Typical activity	Typical activity (2 copies)
SLC18A2	rs363371	G/G	G → Reduced activity	G → Reduced activity	Reduced activity (2 copies)

Gene	SNP	Genotype	1 allele → effect	2 allele → effect	Global activity
HTR3A	rs1062613	T/C	T → Increased activity	C → Typical activity	Mixed: 1 Enhanced Activity + 1 Typical Activity
SLC18A1	rs1390938	G/G	G → Reduced activity	G → Reduced activity	Reduced activity (2 copies)
TPH2	rs4570625	G/T	G → Increased activity	T → Typical activity	Mixed: 1 Enhanced Activity + 1 Typical Activity
DDC	rs921451	T/T	T → Typical activity	T → Typical activity	Typical activity (2 copies)
DDC	rs11575542	C/C	C → Typical activity	C → Typical activity	Typical activity (2 copies)
DDC	rs3735273	C/C	C → Reduced activity	C → Reduced activity	Reduced activity (2 copies)
HTR2A	rs6313	G/G	G → Reduced activity	G → Reduced activity	Reduced activity (2 copies)
HTR2A	rs6311	C/C	C → Reduced activity	C → Reduced activity	Reduced activity (2 copies)
HTR1D	rs6300	G/A	G → Altered activity	A → Typical activity	Mixed: 1 Altered Activity + 1 Typical Activity
HTR1D	rs676643	G/G	G → Typical activity	G → Typical activity	Typical activity (2 copies)

Health theme, Enhancers and Blockers

Gene	SNP	Health theme	Enhancers	Blockers
BDNF	rs6265	support for serotonin signaling, mental resilience and cognition; cognitive problems	exercise, omega-3, curcumin	stress, excess sugar
MAOA	rs6323	monoamine levels	—	—
MAOA	rs909525	monoamine levels	—	—
ALDH2	rs671	elimination of aldehydes and risk of nerve injury	—	—
GSK3B	rs334558	sensitivity of serotonin receptors	magnesium	—
GSK3B	rs6438552	sensitivity of serotonin receptors	magnesium	—
GSK3B	rs3755557	sensitivity of serotonin receptors	magnesium	—
HTR6	rs1805054	risk of mental health disorders; additions	—	—
HTR1A	rs6295	serotonin levels; anxiety, stress, depression and ADHD	magnesium, omega-3, zinc	stress
HTR4	rs1345697	mood	choline	stress
HTR4	rs6889822	mood	choline	stress
HTR2C	rs518147	Altered serotonin signaling	exercise, sleep	excess sugar, processed foods
HTR2C	rs3813929	Altered serotonin signaling	exercise, sleep	excess sugar, processed foods
SLC6A4	rs2129785	recycling and serotonin levels; psychiatric conditions; response to SSRI	—	stress
SLC6A4	rs11867581	recycling and serotonin levels; psychiatric conditions; response to SSRI	—	stress
HTR1B	rs6296	predisposition to pain	magnesium, riboflavin, CoQ10	nitrites/nitrates (processed meat)
HTR1B	rs130060	predisposition to pain	magnesium, riboflavin, CoQ10	nitrites/nitrates (processed meat)
HTR1B	rs11568817	predisposition to pain	magnesium, riboflavin, CoQ10	nitrites/nitrates (processed meat)
SLC18A2	rs363276	serotonin load in the vesicles; PTSD, addition and mental health	—	—
SLC18A2	rs363387	serotonin load in the vesicles; PTSD, addition and mental health	—	—
SLC18A2	rs363371	serotonin load in the vesicles; PTSD, addition and mental health	—	—
HTR3A	rs1062613	Irritable bowel syndrome rates	—	—
SLC18A1	rs1390938	monoamine levels	—	—
TPH2	rs4570625	GABA levels and serotonin production	—	—
DDC	rs921451	L-DOPA metabolism	vitamin B6	—

Gene	SNP	Health theme	Enhancers	Blockers
DDC	rs11575542	L-DOPA metabolism	vitamin B6	—
DDC	rs3735273	L-DOPA metabolism	vitamin B6	—
HTR2A	rs6313	levels and serotonin signaling	—	—
HTR2A	rs6311	levels and serotonin signaling	—	—
HTR1D	rs6300	serotonin signaling	CoQ10, riboflavin, vitamin B6	nitrites/nitrates
HTR1D	rs676643	serotonin signaling	CoQ10, riboflavin, vitamin B6	nitrites/nitrates

Melatonin pathway and circadian rhythm

Melatonin is made from serotonin in the pineal gland (AANAT → ASMT), acts on MT1 (MTNR1A) and MT2 (MTNR1B) receptors, it is eliminated mainly by CYP1A2 and conjugated by SULT1A1. The clock genes (CLOCK, ARNTL/BMAL1, PER2, CRY2) mark the daily rhythm that commands its release. The ASMT is in the pseudoautosomal region of X/Y, which the chip and the imputation do not cover.

Genotype and activity

Gene	SNP	Genotype	1 allele → effect	2 allele → effect	Global activity
MTNR1B	rs10830963	C/G	C → Reduced activity	G → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
ASMT	rs4446909	SNP not found in the data	SNP not found in the data	SNP not found in the data	SNP not found in the data
ASMT	rs5989681	SNP not found in the data	SNP not found in the data	SNP not found in the data	SNP not found in the data
PER2	rs2304672	G/G	G → Increased activity	G → Increased activity	Increased activity (2 copies)
MTNR1A	rs2119882	T/T	T → Higher activity	T → Higher activity	Higher activity (2 copies)
MTNR1A	rs12506228	C/C	C → Higher activity	C → Higher activity	Higher activity (2 copies)
MTNR1A	rs13140012	A/A	A → Lower activity	A → Lower activity	Lower activity (2 copies)
AANAT	rs28936679	G/G	G → Typical activity	G → Typical activity	Typical activity (2 copies)
CLOCK	rs1801260	A/G	A → Reduced activity	G → Increased activity	Mixed: 1 Reduced Activity + 1 Increased Activity
CRY2	rs11605924	C/C	C → Normal activity	C → Normal activity	Normal activity (2 copies)
SULT1A1	rs1042028	C/C	C → Increased activity	C → Increased activity	Increased activity (2 copies)
ARNTL	rs2278749	C/C	C → Typical activity	C → Typical activity	Typical activity (2 copies)
ARNTL	rs2290035	A/A	A → Altered activity	A → Altered activity	Altered activity (2 copies)
ARNTL	rs6486122	T/T	T → Altered activity	T → Altered activity	Altered activity (2 copies)
CYP1A2	rs762551	C/A	C → Reduced activity	A → Typical activity	Mixed: 1 Reduced Activity + 1 Typical Activity

Health theme, Enhancers and Blockers

Gene	SNP	Health theme	Enhancers	Blockers
MTNR1B	rs10830963	delayed release of melatonin, higher blood sugar and risk of diabetes	—	—
ASMT	rs4446909	production of melatonin and 5-HTP	SAMe, magnesium, folate, sleep, vitamin B12	—
ASMT	rs5989681	production of melatonin and 5-HTP	SAMe, magnesium, folate, sleep, vitamin B12	—
PER2	rs2304672	circadian rhythm and melatonin signaling; eating habits and diet adherence	exercise, sleep	alcohol
MTNR1A	rs2119882	melatonin signaling, circadian rhythm, metabolic and cognitive health	Sleep, Omega 3	excess caffeine, alcohol
MTNR1A	rs12506228	melatonin signaling, circadian rhythm, metabolic and cognitive health	Sleep, Omega 3	excess caffeine, alcohol

Gene	SNP	Health theme	Enhancers	Blockers
MTNR1A	rs13140012	melatonin signaling, circadian rhythm, metabolic and cognitive health	Sleep, Omega 3	excess caffeine, alcohol
AANAT	rs28936679	melatonin production; late sleep phase syndrome	Sleep, Omega 3	caffeine in excess
CLOCK	rs1801260	Sleep quality, metabolic health and dietary patterns	—	—
CRY2	rs11605924	control of blood sugar	—	—
SULT1A1	rs1042028	melatonin clearance; mixed effects on detoxification	—	—
ARNTL	rs2278749	circadian rhythm and metabolic health	Omega-3, sleep	stress
ARNTL	rs2290035	circadian rhythm and metabolic health	Omega-3, sleep	stress
ARNTL	rs6486122	circadian rhythm and metabolic health	Omega-3, sleep	stress
CYP1A2	rs762551	melatonin clearance; caffeine sensitivity	riboflavin, iron	—

4. Modified genes and typical genes

Serotonin genes with different results from Typical/Normal

- **HTR2A:** Reduced in 2 SNPs (rs6313 G/G, rs6311 C/C).
- **HTR2C (X):** Reduced in 2 SNPs (rs518147 G, rs3813929 C).
- **HTR1B:** rs130060 Reduced (A/A); rs6296 and rs11568817 Mixed (1 Increased + 1 Reduced).
- **HTR1A:** Mixed (rs6295 C/G: 1 Increased + 1 Reduced).
- **HTR4:** Altered in 2 SNPs (rs1345697 C/C, rs6889822 A/A; imputed, DR2 1.00).
- **HTR3A:** 1 Typical + 1 Increased (rs1062613 T/C).
- **HTR1D:** rs6300 mixed (G/A: 1 Altered + 1 Typical; Imputation); rs676643 Typical.
- **TPH2:** 1 Increased + 1 Typical (rs4570625 G/T).
- **DDC:** rs3735273 Reduced (C/C); rs921451 and rs11575542 Typical.
- **SLC18A1:** Reduced (rs1390938 G/G).
- **SLC18A2:** rs363371 Reduced (G/G); rs363276 and rs363387 Typical (imputed).
- **SLC6A4:** contradictory (rs2129785 T/T → probable "short"; rs11867581 G/G → probable "long").
- **BDNF:** 1 Typical + 1 Reduced (rs6265 C/T, Val66Met).

Genes of melatonin and clock with different result from Typical/Normal

- **MTNR1B:** mixed (rs10830963 C/G: 1 Reduced + 1 Encreaged).
- **MTNR1A:** mixed between SNPs (rs2119882 T/T Higher and rs12506228 C/C Higher; rs13140012 A/A Lower).
- **PER2:** Increased (rs2304672 G/G).
- **CLOCK:** mixed (rs1801260 A/G: 1 Reduced + 1 Encreaged).
- **ARNTL:** 2 Altered (rs2290035 A/A, rs6486122 T/T) + Typical 1 (rs2278749 C/C).
- **SULT1A1:** Increased (rs1042028 C/C; imputed).
- **CYP1A2:** 1 Typical + 1 Reduced (rs762551 C/A). The origin report does not bind this SNP to changes in melatonin levels.

Genes with Typical/Normal Result

- **Serotonin:** MAOA (rs6323 T, rs909525 T; X), ALDH2 (rs671 G/G), GSK3B (3 SNPs), HTR6 (rs1805054 C/C: "Typical HTR6A activity").
- **Melatonin/clock:** AANAT (rs28936679 G/G), CRY2 (rs11605924 C/C: Normal).

Unread

- **ASMT** rs4446909 and rs5989681: SNP not found in the data (X/Y pseudoautosomal region, out of the chip and imputation).

Contradictory or non-concordant SNPs

- **SLC6A4:** the two markers of 5-HTTLPR point in opposite directions. 5-HTTLPR is a repetitive insertion/deletion that the chip does not read; only one directed test solves it.
- **DDC, ARNTL, SLC18A2** and **MTNR1A:** SNPs from the same gene with different tags (it is not an error: they are different sites of the gene).

5. Reading as a system

Read as a whole, without judgment: 1. Serotonin production: the limiting step in the brain (TPH2) has 1 allele Increased; the next step (DDC, shared with dopamine) has 1 SNP reduced in 3. The probable balance is a typical production. 2. Storage and release: the two vesicular transporters have Reduced tags (SLC18A1 and 1 SNP of SLC18A2), which describes a possibly slightly lower vesicular load of monoamines. It is the same SLC18A1 that appears in the dopamine and noradrenaline reports. 3. Reuptake: for clarifying (SLC6A4 contradictory). 4. Signaling: is the most marked point. The post-synaptic receptors HTR2A and HTR2C and the HTR1B are Reduced; the autoreceptor HTR1A is mixed; the HTR4 is Altered. Taken together, it is a profile of serotonergic post-synaptic signaling tending to be lower, without clinical significance by itself. 5. Degradation: typical (MAOA and ALDH2). 6. Melatonin: typical production (AANAT; ASMT per reading), mixed receptors and mixed elimination (CYP1A2 with Reduced 1 allele, SULT1A1 Increased). **MTNR1B G** is the allele in which high melatonin with a meal worsens glucose tolerance, which reinforces taking melatonin away from the last meal. 7. Clock: PER2 Increased, ARNTL Altered and CLOCK mixed: a clock that benefits from regular times of light, sleep and meals. It is in line with the Circadian and Sleep Rhythm report.

Connections to other reports: Dopamine and Noradrenaline (DDC, SLC18A1, MAOA, COMT), Mood and Depression Risk, Anxiety and Stress Resilience, Circadian Rhythm and Sleep, Psychedelic (HTR2A).

6. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Circadian disorders	CLOCK mixed, ARNTL Altered (2 of 3), PER2 Increased, CRY2 Normal	without judgment (descriptive)
Desynchronization of Biological Clocks	Central clock and melatonin signal (MTNR1A mixed, MTNR1B G) sensitive to irregular times	without judgment (descriptive)
Sleep disturbance	Production of typical melatonin (AANAT); mixed receptors; ASMT per reading	without judgment (descriptive)
Hyperglycemia and Intolerance to Glucose	MTNR1B rs10830963 G (1 copy): melatonin and fasting glucose; Normal CRY2	without judgment (descriptive)
Intercellular Communication	Receptors of serotonin HTR2A, HTR2C and HTR1B Reduced; HTR4 Altered	without judgment (descriptive)
Intestinal Barrier dysfunction	~90% of serotonin is intestinal: HTR3A (1 Increased) and HTR4 (Altered) modulate motility	without judgment (descriptive)
Neuroinflammation	BDNFVal66Met heterozygous: plasticity and neuronal resilience	without judgment (descriptive)
Oxidative stress	Melatonin is a mitochondrial antioxidant; ALDH2 Typical (aldehydes)	without judgment (descriptive)
Endocrine dysregulation	Melatonin is the dark hormone; mixed elimination (CYP1A2, SULT1A1)	without judgment (descriptive)

7. Practical suggestions

All suggestions are suggestion (not applied): the real stack of Sérgio F remains as it is, and any change should be discussed with the doctor.

1. **Light in the morning** (exterior, 10–30 min in the first hours) and little light at night: it is the strongest signal for clock genes (ARNTL, CLOCK, PER2) and for the release of melatonin. *Suggestion (not applied).*
2. **Regular times** to lie down, get up and eat, also at the weekend. *Suggestion (not applied).*
3. **Melatonin 0.3–1 mg**, 30–60 min before bedtime and ≥ 3 h after the last meal (MTNR1B G). Larger doses do not bring benefits and prolong the signal in the morning (CYP1A2 with Reduced 1 allele). *Suggestion (not applied).*
4. **Regular exercise** (is Enhancer of BDNF, HTR2C and PER2) and sufficient sleep (Enhancer of HTR2C, MTNR1A, AANAT and ARNTL). *Suggestion (not applied).*
5. **Omega-3 3.2–4 g/day of EPA+DHA** (already in the plan): is Enhancer of BDNF, HTR1A, MTNR1A, AANAT and ARNTL. *Suggestion (not applied).*
6. **Magnesium and zinc** (Enhancers of GSK3B, HTR1A and HTR1B), preferably for food; any supplement only with the doctor. Riboflavin and CoQ10 appear as Enhancers of HTR1B/HTR1D. *Suggestion (not applied).*
7. **Avoid the indicated Blockers:** Chronic stress (BDNF, HTR1A, HTR4, SLC6A4, ARNTL), excess sugar and processed foods (BDNF, HTR2C), meat processed with nitrites (HTR1B, HTR1D), excess or late caffeine (MTNR1A, AANAT) and alcohol (PER2, MTNR1A; is already zero). *Suggestion (not applied).*
8. **Mood during isotretinoin:** maintain the surveillance of mood and sleep already foreseen in the plan. *Suggestion (not applied).*
9. **Serotonergic drugs (if needed):** The response to SSRI is not predictable by these genes (SLC6A4 unexplained). The pharmacogenetics of Sérgio F (CYP2C19 *1/*1, probable normal CYP2D6) allows standard doses. Avoid combining several serotonergic drugs (e.g. tramadol with antidepressants or triptans) without the physician. *Suggestion (not applied).*
10. **Directed 5-HTTLPR test** only if there is a clinical decision that depends on it. *Suggestion (not applied).*

8. Limitations

- The effects of each SNP are small and the tags (Increased, Reduced, Altered...) come from the original report. They do not measure serotonin or melatonin levels.
- **5-HTTLPR** (SLC6A4) is a repetition that the chip does not read; both markers contradict each other.
- **ASMT** is in the pseudoautosomal region of X/Y, outside the chip and imputation.
- Eight SNPs come from imputation (DR2 0.86–1.00); SLC18A2 rs363387 has the lowest DR2 (0.86).
- By Sérgio F rule, there is no favourable/unfavourable judgment in neurotransmitters.
- Conclusions only from genes; it is not a diagnosis.

9. Methods and sources

- **Genes and SNPs:** exactly the 27 genes and 46 SNPs of the original order (31 serotonin, 15 melatonin), with the allele tags of the original report. Nothing was added or replaced.
- **Genotypes:** Sérgio F chip (GRCh38, strand +), reread for this report. Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer; only public positions were consulted by rsID.
- **Guidance:** The alleles of the original report are in the strand + (confirmed by VCF alleles and by the frequency in the reference population), so there was no chain conversion, nor in the 9 palindromic SNPs (G/C).
- **Chromosome X:** MAOA and HTR2C out of pseudoautosomal regions → a letter (hemizygous).
- **Comparison with previous analysis (02-10-2026):** The 36 genotypes of the chip are equal; the 8 imputed confirm the values already integrated in the profile.
- **Context Literature:** GWAS Catalog, dbSNP and reviews on serotonin and melatonin pathway (e.g. MTNR1B and glucose: Lyssenko 2009, Garaulet 2020; 5-HTTLPR: Hu 2006).

10. Annex: all variants analysed

Via	Gene	SNP	Alleles of the report	Genotype	Source
Serotonin pathway	BDNF	rs6265	C=Typical activity/T=Reduced activity	C/T	chip
Serotonin pathway	MAOA	rs6323	G=Increased activity/T=Typical activity	T (hemizygous, X)	chip
Serotonin pathway	MAOA	rs909525	C=Increased activity/T=Typical activity	T (hemizygous, X)	chip
Serotonin pathway	ALDH2	rs671	A=Reduced activity/G=Typical activity	G/G	chip
Serotonin pathway	GSK3B	rs334558	A=Increased activity/G=Typical activity	G/G	chip
Serotonin pathway	GSK3B	rs6438552	A=Increased activity/G=Typical activity	G/G	chip
Serotonin pathway	GSK3B	rs3755557	A=Increased activity/T=Typical activity	T/T	chip
Serotonin pathway	HTR6	rs1805054	C=Typical HTR6A activity/T=Improved HTR6A activity	C/C	chip
Serotonin pathway	HTR1A	rs6295	G=Increased activity/C=Reduced activity	C/G	chip
Serotonin pathway	HTR4	rs1345697	C=Altered activity/T=Normal activity	C/C	imputation DR2=1.00
Serotonin pathway	HTR4	rs6889822	A=Altered activity/G=Normal activity	A/A	imputation DR2=1.00
Serotonin pathway	HTR2C	rs518147	Increased activity	G (hemizygous, X)	chip
Serotonin pathway	HTR2C	rs3813929	C=Reduced activity/T=	C (hemizygous, X)	chip
Serotonin pathway	SLC6A4	rs2129785	C=More likely to carry the "long" 5-HTTLPR allele / T=More likely to carry the "short" 5-HTTLPR allele	T/T	chip
Serotonin pathway	SLC6A4	rs11867581	A=More likely to carry the "short" 5-HTTLPR allele / G=More likely to carry the "long" 5-HTTLPR allele	G/G	chip
Serotonin pathway	HTR1B	rs6296	Increased activity	C/G	chip
Serotonin pathway	HTR1B	rs130060	A=Reduced activity/C=	A/A	chip
Serotonin pathway	HTR1B	rs11568817	A=Reduced activity/C=	A/C	chip
Serotonin pathway	SLC18A2	rs363276	C=Typical activity/T=Reduced activity	C/C	imputation DR2=0.99
Serotonin pathway	SLC18A2	rs363387	G=Increased activity/T=Typical activity	T/T	imputation DR2=0.86
Serotonin pathway	SLC18A2	rs363371	A=Increased activity/G=Reduced activity	G/G	chip
Serotonin pathway	HTR3A	rs1062613	C=Typical activity/T	T/C	chip
Serotonin pathway	SLC18A1	rs1390938	A=Increased activity/G=Reduced activity	G/G	chip
Serotonin pathway	TPH2	rs4570625	G=Increased activity/T=Typical activity	G/T	chip
Serotonin pathway	DDC	rs921451	C=Reduced activity/T=Typical activity	T/T	chip
Serotonin pathway	DDC	rs11575542	C=Typical activity/T=Reduced activity	C/C	chip
Serotonin pathway	DDC	rs3735273	C=Reduced activity/T=	C/C	chip
Serotonin pathway	HTR2A	rs6313	A=Increased activity/G=Reduced activity	G/G	chip
Serotonin pathway	HTR2A	rs6311	C=Reduced activity/T=	C/C	chip
Serotonin pathway	HTR1D	rs6300	A=Typical activity/G=Altered activity	G/A	imputation DR2=1.00
Serotonin pathway	HTR1D	rs676643	A=Altered activity/G=Typical activity	G/G	chip
Melatonin pathway and circadian rhythm	MTNR1B	rs10830963	G=Increased activity/C=Reduced activity	C/G	chip
Melatonin pathway and circadian rhythm	ASMT	rs4446909	A=Increased activity/G=Reduced activity	SNP not found in the data	SNP not found in the data
Melatonin pathway and circadian rhythm	ASMT	rs5989681	Increased activity	SNP not found in the data	SNP not found in the data
Melatonin pathway and circadian rhythm	PER2	rs2304672	G=Increased activity/C=Reduced activity	G/G	chip

Via	Gene	SNP	Alleles of the report	Genotype	Source
Melatonin pathway and circadian rhythm	MTNR1A	rs2119882	C=lower activity/T=Higher activity	T/T	imputation DR2=0.93
Melatonin pathway and circadian rhythm	MTNR1A	rs12506228	A=lower activity/C=Higher activity	C/C	chip
Melatonin pathway and circadian rhythm	MTNR1A	rs13140012	A=lower activity/T=Higher activity	A/A	imputation DR2=1.00
Melatonin pathway and circadian rhythm	AANAT	rs28936679	A=Altered activity/G=Typical activity	G/G	chip
Melatonin pathway and circadian rhythm	CLOCK	rs1801260	A=Reduced activity/G=Increased activity	A/G	chip
Melatonin pathway and circadian rhythm	CRY2	rs11605924	A=Altered activity/C=Normal activity	C/C	chip
Melatonin pathway and circadian rhythm	SULT1A1	rs1042028	C=Increased activity/T=Reduced activity	C/C	imputation DR2=0.98
Melatonin pathway and circadian rhythm	ARNTL	rs2278749	C=Typical activity/T=Altered activity	C/C	chip
Melatonin pathway and circadian rhythm	ARNTL	rs2290035	A=Altered activity/T=Normal activity	A/A	chip
Melatonin pathway and circadian rhythm	ARNTL	rs6486122	C=Normal activity/T=Altered activity	T/T	chip
Melatonin pathway and circadian rhythm	CYP1A2	rs762551	A=Typical activity/C=Reduced activity	C/A	chip